

P06. Eco-driving for Electric Connected Vehicles at Signalized Intersections: A Parameterized Reinforcement Learning Approach

中英双语博士精读笔记

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Theme / 主题: Electric CV eco-driving + parameterized RL

Method family / 方法族: Parameterized reinforcement learning / 参数化强化学习

PDF pages / 页数: 27 **Relevance / 相关度:** 83/100

Source / 来源: <https://arxiv.org/abs/2206.12065>

1 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

1.1 研究背景与痛点 / Background & Pain Points

这篇论文位于“Electric CV eco-driving + parameterized RL”方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of Electric CV eco-driving + parameterized RL. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Pain point / 痛点: Signalized intersections and electric connected vehicles rather than fully signal-free autonomous scheduling.

1.2 核心科学问题 / Core Scientific Question

如何同时处理离散驾驶决策和连续控制参数，使电动网联车在信号交叉口实现节能通行？

How can discrete maneuver choices and continuous control parameters be learned jointly for EV eco-driving at signalized intersections?

1.3 科学贡献 / Scientific Contributions

- 方法贡献 (method contribution): Parameterized RL combines model-based car-following, lane-changing policy, and learned longitudinal/lateral decisions.

- 安全/避碰贡献 (safety contribution): Safety is handled through model-based car-following and lane-changing constraints around human-driven vehicles.
- 平滑/停止避免贡献 (smoothing contribution): Learns energy-saving actions around signalized intersections without interrupting surrounding HDVs.
- 创新力度评判 (reviewer judgement): 中等偏强: 参数化动作空间贴合驾驶决策结构, 对混合离散-连续调度有启发。

1.4 核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。
Parameterized reinforcement learning	参数化强化学习	该论文的核心方法族。

2 教材式学习路径与关键截图 / Textbook-Style Study Path & PDF Snapshots

2.1 学习路径 / Study Path

- 第一遍先读首页截图、摘要与引言, 确认研究问题和场景假设。
- 第二遍沿着技术路线图复现模型变量、约束和求解器输入输出。
- 第三遍逐节阅读 section deep reading, 对照 PDF 截图检查公式、图表和实验。
- 第四遍只站在审稿人视角重读实验, 判断 baseline、公平性、消融和鲁棒性。
- 最后把博士 checklist 写成自己的复现实验或论文拓展计划。

2.2 博士阅读 Checklist / Doctoral Reading Checklist

- 能否独立重写核心公式: 参数化动作 MDP / Parameterized-action MDP, 并解释每个符号的物理意义?
- 能否画出输入-模型-算法-控制-验证的数据流图?
- 能否指出至少两个强假设, 并说明它们在真实交通中何时失效?
- 能否复述实验基准是否公平, 指标是否覆盖效率、安全、能耗和计算时间?
- 能否提出一个与自己 JSSP/RL/smoother 课题直接相连的拓展实验?

2.3 关键截图讲解 / PDF Snapshot Explanation

Screenshots are selected anchors from the local PDFs for study and parallel reading. They do not replace the original paper.

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ORIGINAL ARTICLE

Eco-driving for Electric Connected Vehicles at Signalized Intersections: A Parameterized Reinforcement Learning approach

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ARTICLE HISTORY

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ABSTRACT

This paper proposes an eco-driving framework for electric connected vehicles (CVs) based on reinforcement learning (RL) to improve vehicle energy efficiency at signalized intersections. The vehicle agent is specified by integrating the model-based car-following policy, lane-changing policy, and the RL policy, to ensure safe operation of a CV. Subsequently, a Markov Decision Process (MDP) is formulated, which enables the vehicle to perform longitudinal control and lateral decisions, jointly optimizing the car-following and lane-changing behaviors of the CVs in the vicinity of intersections. Then, the hybrid action space is parameterized as a hierarchical structure and thereby trains the agents with two-dimensional motion patterns in a dynamic traffic environment. Finally, our proposed methods are evaluated in SUMO software from both a single-vehicle-based perspective and a flow-based perspective. The results show that our strategy can significantly reduce energy consumption by learning proper action schemes without any interruption of other human-driven vehicles (HDVs).

KEYWORDS

eco-driving, reinforcement learning, connected vehicles, signalized intersections

1. Introduction

Transportation system has been recognised as one of the major sources of energy consumption and air pollution. The authorities have tried their utmost to build a sustainable and efficient transportation system by applying advanced technologies. One promising way to facilitate the progress is to increase the market penetrating rate (MPR) of electric vehicles with effective energy management strategies. On one hand, the electrification of vehicles promote the use of cleaner energy, which is of benefit to achieve low-emission outcomes (Li et al. 2019). On the other hand, making full use of existing energy can help reduce energy consumption during the operation period of the vehicles. As an energy efficient purpose is inseparable from advanced technologies, the application of vehicle-to-everything (V2X) communication to traffic control and management have triggered a possible revolution towards an ecological and efficient transportation system. A plenty of research-based evidences can be found to support the view that such connected technologies are promising to enhance traffic performance (Jia and Ngoduy 2016; Talebpour and Mahmassani 2016; Ard et al. 2021; Shi et al.

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图 1: 首页与摘要 / Title and abstract page (p.1, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Provides a SUMO-compatible example of energy and stop reduction metrics for the experimental section.

former element is expected to be processed by a convolutional neural network block to extract latent features, and the latter element will concatenate to the extracted local traffic state features and be input to a multi-layer perceptron block to train the PRL policy.

Secondly, the action space in this paper is considered as a hybrid form that support both longitudinal control and lateral decision. As mentioned above, the basic representation of the action at time t is $a_t = (k_t, x_{k_t})$, where $k_t \in \{-1, 0, 1\}$ and $x_{k_t} \in [a_{min}, a_{max}]$. The value set $\{-1, 0, 1\}$ of k_t represent that the ego vehicle can change to the left lane, stay at the current lane, and change to the right lane. We assume that the lane changing process can be finished in a fixed duration T_c . The space of x_{k_t} limits the acceleration of the CV to the range of its maximal deceleration a_{min} and maximal acceleration a_{max} . Instead of combining discrete speed variation rate and lane changing decision to formulate an overall discrete action space, we utilize the hybrid discrete-continuous action space to effectively explore the solution space and obtain better performance. Since imposing the original a_t on the ego vehicle may make it be exposed to traffic accident, we introduce conventional car-following model and lane changing model to modify the dangerous driving behavior. In this paper, we utilize the Krauss car-following model (Krauß 1998) and the "LC2013" lane-changing model (Erdmann 2015), which are embedded in SUMO simulation software, to perform clip and mask operations and obtain the modified action $\hat{a}_t = (\hat{k}_t, \hat{x}_{\hat{k}_t})$:

- Clip operation aims at clipping the potential risking acceleration to a safe value to avert rear-end collisions or beyond the road speed limit V_{max} . We assume that the acceleration output by the utilized car-following model is x_t^{CF} at time t according to current driving status, and then the corrected acceleration is transformed to $\hat{x}_{k_t} = \min(x_t^{CF}, x_{k_t})$. Hence, the velocity of agent i at time t is clipped as $v_t^i = \max(\min(V_{max}, v_{t-1}^i + \hat{x}_{k_t}), 0)$, which confines the velocity of the CV to a reasonable range.
- Mask operation is used to judge whether the lateral decision produced by PRL policy is safe, in order to prevent lateral collisions or deadlocks. For each non-zero discrete action k_t , the model-based lane changing policy output an bool value ξ to resolve that if the agent could change lane. The corrected lateral action is denoted as $\hat{k}_t = k_t \times \xi$.

Thirdly, we discuss different settings of the reward function r_t , while different reward signal may trigger various learning directions of the agent. The general goal of the eco-driving strategy is to make a trade-off between energy consumption and travel time, as the energy-efficient driving mode may cause the increase of the delay (Wang et al. 2020). An intuitive way to set the reward function is simply returning a linear combination of total electricity consumption E_f and travel time T_f :

$$r_t^1 = \begin{cases} -(E_f + \varphi T_f), & \text{if } t = t_f \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where, φ is a weighting parameter to measure the relative importance of energy indicator and mobility indicator; t_f is the terminal time step (i.e., the time that the ego vehicle leaves the junction).

However, the travel time can only be calculated at the end of the travel, and it is difficult for the vehicle agent to learn effectively from such a delay reward signal.

图 2: 建模或问题设定页 / Modeling or problem-setup page (p.9, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Provides a SUMO-compatible example of energy and stop reduction metrics for the experimental section.

the former estimates Q^{π^*} and generates actions in a greedy way, while the latter directly learns the optimal policy π^* by optimizing $J(\pi)$. As one typical representative of value-based methods, the Q-learning algorithm (Watkins and Dayan 1992) is deduced by Bellman equation:

$$Q(s, a) = \mathbb{E}_{r_t, s_{t+1}} [r_t + \gamma \max_{a' \in \mathcal{A}} Q(s_{t+1}, a') | s_t = s, a_t = a] \quad (1)$$

which takes Q^{π^*} as the unique solution. The learning process iteratively updates the function $Q(s, a)$ through Monte-Carlo method over a series of sample transitions. Conventionally, $Q(s, a)$ is stored in tabular data structure for finite state space configuration. Nevertheless, the problem of large state space arises along with the increase of the task complexity. When the state space is too large to be stored in computer memory or Q -value cannot be retrieved in real time, the function approximation approach is introduced, especially promoted by the development of deep neural networks. DQN utilizes a deep neural network $Q(s, a; \theta) \approx Q(s, a)$ with parameter set θ to approximate Q^{π^*} (Mnih et al. 2015). The learning of the agent is fundamentally based on the gradient descent theory, which takes the following loss function:

$$L_t(\theta) = \{Q(s_t, a_t; \theta) - [r_t + \gamma \max_{a' \in \mathcal{A}} Q(s_{t+1}, a'; \theta_t)]\}^2 \quad (2)$$

Differing from the value-based approaches, the policy-based algorithms with deep neural network directly model the parametrized policy π_θ . The parameter set θ is updated iteratively via gradient descent to maximize $J(\pi_\theta)$. A typical application is REINFORCE algorithm, which takes the corresponding loss function:

$$\Delta_\theta J(\pi_\theta) = \mathbb{E}_{s_t, a_t} [\Delta_\theta \log \pi_\theta(a_t | s_t) Q^{\pi_\theta}(s_t, a_t)] \quad (3)$$

3. Methodology

3.1. Scenario description

In this paper, we consider a fully electric vehicle environment to better measure the holistic energy performance of the mixed road traffic, which is composed of HDVs and CVs in urban scenario, while only CVs can receive the SPaT data and be controlled by eco-driving controller. By incorporating such a mixed traffic condition, the developed strategy is closer to the urban traffic in the near future so that it can be more practical.

Figure 1 shows the general eco-driving process of the ego vehicle. The left half of the figure depicts a typical longitudinal trajectory and speed profile of a CV with eco-driving system. Compared with HDVs, the CV with eco-driving controller can generate more smooth speed profile and navigate the vehicle cross the junction in green phase, in order to minimize its electricity consumption. Considering a more specific scenario, which is shown in the right half of Figure 1, the CV on the road can constantly detect the surrounding traffic state within its perceived range. Inspired by (Wang et al. 2021), we define the perceived range of the CV as a two-dimensional occupancy grid. The grid cells reflect the occupancy situation of corresponding road segments around the CV, so we call the grid local traffic state. Whereas the CVs are equipped with multiple sensor devices, such as camera, Lidar, and radar, this process can be certainly achieved. In addition to local traffic state, the CV could also acquire the SPaT data based on V2I

图 3: 核心方法或算法页 / Core method or algorithm page (p.6, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Provides a SUMO-compatible example of energy and stop reduction metrics for the experimental section.

the convolutional calculation. In addition to the convolutional layers, the kernel sizes of max-pooling layers are all set to 2 with a 2-step stride, while an one-dimension padding operation is also attached to the second max-pooling layer. The configuration of layer "Conv.2.1" and "Conv.2.2" is the same as those in action parameter network. The number of neurons for "Dense.1.1" and "Dense.1.2" is set to 128 and 64, and for "Dense.2.1" and "Dense.2.2" is 256 and 64.

Table 2.: Hyperparameters of the PRL algorithm

Hyperparameter	Symbol	Value
Discount factor	γ	0.99
Averaging rate for Q	τ_1	10^{-2}
Averaging rate for μ	τ_2	10^{-3}
Step size for Q	α	10^{-4}
Step size for μ	β	10^{-5}
Initial exploration factor	ϵ_{init}	1
Terminal exploration factor	ϵ_{end}	10^{-2}
The number of decaying episodes	n_{step}	10^3
Minibatch size	B	128
Replay buffer size	$\mathcal{R}$	5×10^5

5. Simulation and discussion

5.1. Simulation configuration

The simulation environment is built in SUMO (Lopez et al. 2018) to test the performance of both training phase and evaluation phase. As an highly portable microscopic traffic simulation platform, SUMO offers abundant pre-defined driver models, and meanwhile the operation of vehicles can be controlled through Traci interface by Python programs. Therefore, we delimit the HDVs and CVs by adding PRL policy programmed in Python module to CVs, while the operation of HDVs is based on the default Krauss car-following model and LC2013 lane-changing model. The energy model that produces transient electricity consumption is derived from (Kurczveil, López, and Schnieder 2013), which leverages a braking energy recovery mechanism. It means that the value of instantaneous energy consumption e_t^i is negative when the vehicle perform a braking operation, which presents challenges to searching a globally optimal eco-driving policy. The specific form and parameters of the energy model are given in our previous article (Zhang et al. 2022). The proposed PRL policy do not need to explicitly analysis the energy model, while it only receive a transient signal that indicates the energy consumption in a model-agnostic manner.

We create a coordinated signal control scenario and a non-coordinated signal control scenario in a multi-intersection environment to evaluate the proposed framework. As illustrated in Figure 5, the corridor contains five consecutive signalized intersections, for which a V2I communication range is set to 300m. The traffic lights with cycle time of 90s change synchronously for non-coordinated scenario, while for coordinated scenario a series of offset is set up to ensure that a vehicle can encounter a green wave under free flow condition.

The parameters related to the simulation follow those in Table 3 if no otherwise specified. Besides, the values of the penalty items is set as $r_p^s = -30$, $r_p^j = -30$, and $r_p^c = -50$. Together with the weighting parameter φ , the reward-related parameters

图 4: 实验或结果页 / Experiments or results page (p.15, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Provides a SUMO-compatible example of energy and stop reduction metrics for the experimental section.

2021; Jiang et al. 2022). Based on the belief that the communication modules will be deployed simultaneously on both vehicles and traffic infrastructures, the vehicle-to-infrastructure (V2I) technique can be widely implemented in the near future, which enables the connected vehicles (CVs) to obtain the information transmitted from traffic infrastructures such as traffic lights and ramp meterings, with the intent to help CVs make better decisions to improve energy efficiency.

One of a successful applications of V2I communication in urban traffic is the deployment of signal phase and timing (SPaT) message, furnishing CVs with signal status information in a consistent manner (Mintsis, Vlahogianni, and Mitsakis 2020). Since traffic lights in urban signalized intersections can interrupt the operation of vehicles, CVs may avoid the red phases and achieve energy-efficient driving through the use of real-time SPaT data, while idling at the traffic signals is one major cause of increased energy consumption and greenhouse gas emissions (Kim and Kim 2020). Given this, the integration of SPaT information and advanced control policies initiate eco-driving concept, which has potential to significantly reduce energy consumption in the proximity of signalized intersections (Li et al. 2018). Generally, the control policies take vehicle location, SPaT data, and local traffic state as input and generate speed profile for the controlled vehicle (i.e., the ego vehicle), and thus enhance the performance of the vehicle in terms of energy conservation (Asadi and Vahidi 2011; Mintsis et al. 2021; Dong et al. 2021; Ma et al. 2021).

Eco-driving strategy at signalized intersections is usually accomplished by keeping up a smooth driving speed, refraining from aggressive acceleration, and avoiding long-time idling at the intersections, resulting in higher performance in energy economy and travel efficiency (Gao et al. 2019; Typaldos, Papamichail, and Papageorgiou 2020). By drawing upon the V2I communication, relevant strategies place value on the design of control algorithms in assorted traffic contexts, aiming at explicitly optimize the energy-saving speed profiles of vehicles. As such, the existing eco-driving strategies can be fundamentally classified into three broad categories: rule-based strategy, optimization-based strategy, and learning-based strategy. The mathematical formulation of each kind of strategy is different, but energy-saving performances with varying degrees are reported for all of the above methods.

Predefined rules are provided for a CV in rule-based strategies when the ego vehicle approaching the intersection. The basic ideas of rule-based methods are similar: they all divided the state into several categories according to the signal status and vehicle status, provided that the SPaT messages are available from V2I communication (Rakha and Kamalanathsharma 2011; Xia, Boriboonsomsin, and Barth 2013; Ci et al. 2019). As a result, the trajectory of the ego vehicle can be simply planned with classical kinematics formulas, trying to let the vehicle cross the junction within green phase.

An advantage of the rule-based methods is that they are easy to be implemented. However, the approaches cannot adapt to dynamic traffic, because it is difficult to take the surrounding vehicles into account. For example, if the vehicle ahead of the ego CV slow down unexpectedly or the vehicle in adjacent lane insert to the placement in front of the ego CV, the rule-based methods may lose their efficacy, whereas some of them only consider the signal status and the vehicle status. Moreover, the rule-based nature is prone to lead local optimum as the speed change rate is usually fixed. In this case, the results can leave much to be desired.

In order to achieve better performance, the optimization-based approaches are extensively studied. One representative way in optimization-based strategies is to formulate an optimal control model and generate corresponding speed profiles, which is

图 5: 补充精读页 / Supplemental close-reading page (p.2, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Provides a SUMO-compatible example of energy and stop reduction metrics for the experimental section.

3 章节精读与逐段导读 / Section-by-Section Deep Reading & Walkthrough

p.1 1. Introduction

章节目标 / Learning goal: 建立研究动机：为什么传统信号控制、FCFS 或单车控制不足以处理该论文关注的交叉口协同问题。

Understand why the paper needs a new coordination method rather than a standard signal, FCFS, or single-vehicle controller.

关键论点 / Key argument: 这一部分通常把痛点落到 Electric CV eco-driving + parameterized RL：既要提高效率，又要保持安全与在线可执行。

技术焦点 / Technical focus: 精读时标出作者批判的 baseline、场景假设、交通参与者类型和隐含优化目标。

审稿追问 / Reviewer question: 作者指出的痛点是否真的需要新方法，还是可以由更强的优化器/更保守规则解决？

p.5 2. Background of Reinforcement Learning

章节目标 / Learning goal: 建立方法谱系和相关工作边界，避免把本文贡献误读为完全从零开始。

Place the paper inside the broader method taxonomy.

关键论点 / Key argument: 该部分说明作者站在哪些已有工作之上，以及本文选择绕开或继承了哪些限制。

技术焦点 / Technical focus: 把相关工作按 scheduler、smoother、safety filter、evaluation 四类重排。

审稿追问 / Reviewer question: 作者是否公平比较了最接近的强 baseline？

p.6 3. Methodology

章节目标 / Learning goal: 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 参数化 RL 把驾驶行为拆成离散模式 and 对应连续参数，结合模型式跟驰/变道规则处理周围 HDV，再用学习策略选择节能动作。该结构很适合未来把高层调度选择和低层连续轨迹合并。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.6 3.1. Scenario description

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.7 3.2. Agent framework

章节目标 / Learning goal: 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 参数化 RL 把驾驶行为拆成离散模式 and 对应连续参数，结合模型式跟驰/变道规则处理周围 HDV，再用学习策略选择节能动作。该结构很适合未来把高层调度选择和低层连续轨迹合并。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.10 4. PRL for eco-driving

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.10 4.1. Reinforcement Learning in Hybrid Action Space

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.12 4.2. PRL algorithm

章节目标 / Learning goal: 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 参数化 RL 把驾驶行为拆成离散模式 and 对应连续参数，结合模型式跟驰/变道规则处理周围 HDV，再用学习策略选择节能动作。该结构很适合未来把高层调度选择和低层连续轨迹合并。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

3.1 逐段式导读 / Paragraph-Level Walkthrough

Opening motivation / 开篇动机段 先把作者的痛点翻译成一句博士问题：如何同时处理离散驾驶决策和连续控制参数，使电动网联车在信号交叉口实现节能通行？读这一段时不要急着接受动机，要追问传统 FCFS、信号灯或保守 MPC 为什么不足。

Turn the opening motivation into one doctoral-level question and test whether the paper truly needs a new method.

Assumption and setting / 假设与场景段 把车辆类型、通信条件、路口类型和动力学假设列成表。该文的关键边界包括：周围车辆行为可由跟驰/变道模型近似。；电动车能耗模型可靠，且可用于仿真奖励。；信号灯信息和车辆状态可观测。

List vehicle type, communication, intersection setting, and dynamics assumptions before reading the equations.

Model and notation / 模型与符号段 围绕核心公式“参数化动作 MDP / Parameterized-action MDP”复写符号表，确认每个变量是决策变量、状态变量、参数还是约束阈值。

Rewrite the symbol table around the core formulation and classify variables as states, decisions, parameters, or constraints.

Algorithm mechanism / 算法机制段 把算法读成一条数据流：输入交通状态，执行 Parameterized RL combines model-based car-following, lane-changing policy, and learned longitudinal/lateral decisions.，输出可执行通行/控制决策，并检查安全接口。

Read the algorithm as a dataflow from traffic state to executable sequence/control decision, with explicit safety interfaces.

Experiment and evidence / 实验证据段 不要只看作者报告的提升，要检查基准、公平性和指标。本文验证描述为：SUMO evaluation from single-vehicle and flow perspectives.

Go beyond reported gains and inspect baselines, fairness, metrics, and computation time.

Reviewer synthesis / 审稿式总结段 最后把贡献拆成可发表与需补强两部分。对你课题最直接的用途是：Provides a SUMO-compatible example of energy and stop reduction metrics for the experimental section.

Separate publishable contribution from missing evidence, then connect the result to your own thesis architecture.

4 技术路线图与贡献量评估 / Technical Route & Contribution Matrix

Step	Stage / 阶段	Content / 内容
1	输入 / Input	车辆状态、路径/车道、交通需求、信号或协同信息。
2	建模 / Modeling	参数化动作 MDP / Parameterized-action MDP

3	核心求解 / Core solver	Parameterized RL combines model-based car-following, lane-changing policy, and learned longitudinal/lateral decisions.
4	安全接口 / Safety interface	Safety is handled through model-based car-following and lane-changing constraints around human-driven vehicles.
5	平滑与效率 / Smoothing and efficiency	Learns energy-saving actions around signalized intersections without interrupting surrounding HDVs.
6	验证 / Validation	SUMO evaluation from single-vehicle and flow perspectives.

Dimension / 维度	Score	Comment / 评语
理论贡献 / Theoretical contribution	3/5	参数化动作 MDP / Parameterized-action MDP
算法贡献 / Algorithmic contribution	4/5	Parameterized RL combines model-based car-following, lane-changing policy, and learned longitudinal/lateral decisions.
实验贡献 / Experimental contribution	3/5	SUMO evaluation from single-vehicle and flow perspectives.
工程贡献 / Engineering contribution	3/5	Safety is handled through model-based car-following and lane-changing constraints around human-driven vehicles.
博士课题价值 / PhD relevance	4/5	Provides a SUMO-compatible example of energy and stop reduction metrics for the experimental section.

5 理论方法论深度解构 / Methodology & Theoretical Framework

5.1 问题数学建模 / Mathematical Formulation

参数化动作 MDP / Parameterized-action MDP

$$a_t = (d_t, \xi_t), \quad d_t \in \mathcal{D}, \quad \xi_t \in \mathbb{R}^{m(d_t)}, \quad Q(s_t, d_t, \xi_t) \approx \mathbb{E}[R_t]$$

Symbol / 符号	含义	Meaning	Role / 建模作用
d_t	离散驾驶决策	discrete driving decision	如保持、变道、跟驰模式
ξ_t	连续动作参数	continuous action parameter	如目标速度或加速度幅值
$\mathcal{D}$	离散动作集合	discrete action set	描述可选机动
Q	动作价值函数	action-value function	评价离散-连续组合收益

R_t	回报	return	累积能耗、效率和安全目标
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5.2 核心算法架构 / Core Algorithm Layout

参数化 RL 把驾驶行为拆成离散模式 and 对应连续参数, 结合模型式跟驰/变道规则处理周围 HDV, 再用学习策略选择节能动作。该结构很适合未来把高层调度选择和低层连续轨迹合并。

Parameterized RL separates maneuver selection from continuous parameter control, combining learned decisions with model-based car-following and lane-changing constraints.

5.3 收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility

模型式规则给安全留出底线, 学习器主要优化效率和能耗; 但混合动作空间训练难度较高, 参数边界设计会影响收敛。

Model-based constraints provide a safety floor, while the learner optimizes energy and efficiency; convergence depends strongly on action parameterization.

5.4 潜在假设与边界条件 / Assumptions & Boundary Conditions

- 周围车辆行为可由跟驰/变道模型近似。
- 电动车能耗模型可靠, 且可用于仿真奖励。
- 信号灯信息和车辆状态可观测。

6 实验验证与结果评判 / Experimental Validation & Result Evaluation

Baselines & Environment / 基准与环境: 传统生态驾驶、普通 DQN/DDPG、规则驾驶、无参数化 RL 和 IDM。

Validation / 验证: SUMO evaluation from single-vehicle and flow perspectives.

Metrics / 指标: 电耗、平均速度、延误、停车次数、舒适性和混合交通影响。

Ablation / 消融: 应比较参数化 RL 与纯离散、纯连续动作策略; 还应测试参数边界和模型式安全规则的贡献。

Reviewer reading / 审稿式阅读提醒: 阅读实验时不要只看平均值; 应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

6.1 图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 33 figure references and 8 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
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系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。
策略网络/训练流程图	policy-network or training-pipeline diagram	检查状态、动作、奖励、经验回放和安全惩罚是否闭环一致。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

7 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

7.1 论文局限性 / Limitations & Weaknesses

- 信号交叉口和电动车设定使结论不完全适用于无信号 CAV 调度。
- 参数化动作设计依赖专家经验。
- 混合交通中 HDV 模型错误可能传播到策略学习。

7.2 博士课题拓展点 / Research Extensions for PhD

- 把 JSSP 的车辆排序作为离散动作，把目标时间/速度作为连续参数。
- 加入不确定 HDV 模型集合，做鲁棒参数化 RL。
- 用分层策略把 lane choice、arrival time 和 speed profile 分别建模。

7.3 如何服务当前课题 / Use in This Project

Provides a SUMO-compatible example of energy and stop reduction metrics for the experimental section.

8 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数 / Parameter: CAV 渗透率 / CAV penetration rate, range [0.05, 1.0] .

渗透率提高通常增强节能协同，但低渗透率下收益受 HDV 约束明显。

Higher penetration improves eco-coordination, while low penetration is constrained by HDVs.

8.1 HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar：题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator，右侧为五大精读模块。
- 公式卡片使用 `<pre>` 或 LaTeX block，紧跟符号表。
- 审稿人批注用 reviewer-note block，博士拓展用 research-idea list。

- 交互参数用 `input[type=range]` + canvas 曲线，打印 PDF 时隐藏交互控件但保留解释。

9 来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

Page	Extracted heading
p.1	1. Introduction
p.6	3. Methodology
p.15	5. Simulation and discussion

p.1 1. Introduction 定位问题背景、研究缺口和为什么现有保守/非协同方法不足。/ Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.5 2. Background of Reinforcement Learning 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.6 3. Methodology 给出算法流程和求解机制，应重点追踪输入、输出和安全接口。/ Gives the algorithmic mechanism; track inputs, outputs, and safety interfaces.

p.6 3.1. Scenario description 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.7 3.2. Agent framework 给出算法流程和求解机制，应重点追踪输入、输出和安全接口。/ Gives the algorithmic mechanism; track inputs, outputs, and safety interfaces.

p.10 4. PRL for eco-driving 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.10 4.1. Reinforcement Learning in Hybrid Action Space 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.12 4.2. PRL algorithm 给出算法流程和求解机制，应重点追踪输入、输出和安全接口。/ Gives the algorithmic mechanism; track inputs, outputs, and safety interfaces.